

MUHAMMAD FARHAT MEHDI

AI AUTOMATION • ROBOTICS & EDGE AI • MACHINE LEARNING • AGENTIC / VOICE AI

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PROFESSIONAL SUMMARY

Computer Engineering student at UET Taxila building end-to-end intelligent systems across AI automation, robotics, machine learning, agentic AI, LLM/RAG, voice systems, and embedded/edge AI. Hands-on work spans ROS2/Jetson robotics, browser automation, adaptive AI agents, ML pipelines, CRM/workflow automation, and deployed full-stack AI products. Key evidence includes ~3-4 s warm-response latency in a conversational robotic proxy, Precision@50 improvement from 0.48 to 0.66 in a FlyRank ML capstone, and WatchMyWork - a deployed adaptive browser-automation platform.

TECHNICAL SKILLS

Programming & Data	Python, C++, Java, SQL, MATLAB
AI / ML / LLMs	Machine Learning, Generative AI, RAG, Hugging Face, FAISS, QLoRA/LoRA, Whisper, XTTS-v2
Agentic AI & Automation	n8n, GoHighLevel, Make, Zapier, Vapi, Retell, Apollo.io, Playwright, Chrome Extensions, Tavily, OpenRouter
Robotics & Edge AI	ROS2, Nav2/SLAM, Gazebo, NVIDIA Jetson/Xavier NX/Nano, LiDAR, GPS/encoders, Edge Impulse
Computer Vision	OpenCV, YOLO/perception monitoring, image-processing workflows
Development & Infra	FastAPI, React, TypeScript, PostgreSQL, SQLite, Git/GitHub, Linux, Docker, Vivado, Cadence

EXPERIENCE & INTERSHIPS

Freelance AI Automation Developer GoHighLevel, n8n, Make, Zapier	Current
Design and build AI-enabled CRM/workflow automations for lead handling, voice-agent workflows, process streamlining, and business operations using Python, webhooks, Vapi, Retell, and related integrations.	
Robotics Intern BrainSwarm Robotics Lab	8 weeks
Machine Learning Intern FlyRank	Completed
Generative AI Intern Aspire Pakistan	Completed
Generative AI Intern Arch Technologies	6 weeks
Voice Agent Project Contributor Startup Project	~2 months
Contributed workflow logic and automation development for a free-API-based voice-agent project.	

FEATURED PROJECT

WatchMyWork - Agentic Browser Automation Platform React • FastAPI • Playwright • Chrome Extension • NVIDIA Nemotron • Tavily • SQLite • Railway
- Built a deployed system that learns a browser workflow once and executes it deterministically across structured data. - Developed an Adaptive Recovery Agent that detects website changes, finds equivalent actions (e.g., Login → Sign In), explains reasoning/confidence, requests approval, and resumes execution. - Integrated IBM Bob Debug Package support for the IBM Bob 2.0 Hackathon and later extended the product with search/recovery capabilities for agentic-AI workflow automation.

FEATURED FINAL YEAR PROJECT

AI-Powered Robotic Proxy for Remote Business Presence in Expos | ROS2 • NVIDIA Jetson • Whisper • RAG/LLM • XTTS-v2 • React

- Built a live speech-to-response robotics pipeline integrated with teleoperation, camera/perception monitoring, transcripts, system status, safety controls, and latency tracking.
- Improved warm conversational response latency to approximately 3-4 s; ongoing work includes LiDAR/navigation, SLAM/Nav2, dedicated robot audio, and simultaneous movement/perception/voice testing.

SELECTED PROJECTS

RoboOps AI - Autonomous Robot Operations Platform | ROS2 • Nav2 • Gazebo • OpenCV • FastAPI • React • PostgreSQL

- End-to-end robot-operations platform connecting local AI mission planning, human approval, ROS2/Nav2/Gazebo, computer vision, live telemetry, n8n automation, PostgreSQL, and automatic PDF mission reporting.

FlyRank ML Capstone | Python • Logistic Regression • Random Forest • Gradient Boosting

- Built an end-to-end ML pipeline on pseudonymized search data; Logistic Regression improved Precision@50 from 0.48 to 0.66 (+37.5% relative).

UGV Mobile Robot & Edge AI | Jetson Nano • LiDAR • GPS • Encoders • ROS2 • Edge Impulse

- Integrated mobile-robot sensing, embedded motor control, dashboard monitoring, and edge-inference workflows using sensor data.

EDUCATION

B.Sc. Computer Engineering | University of Engineering and Technology (UET), Taxila

2023-2027 | CGPA: 3.31/4.00 (through 6th semester)

- Relevant coursework: Embedded Systems & Design, Signals & Systems, Digital Signal Processing, Linear Algebra, Programming Fundamentals (C++), Data Structures & Algorithms (Python), Probability & Statistics, Computer Organization & Architecture.

AWARDS, HACKATHONS & PROGRAMS

- First Runner-Up - Inter-University National AI Hackathon (AUREX 2026).
- IBM Bob 2.0 Hackathon - participated and submitted WatchMyWork with IBM Bob integration; Certificate of Participation.
- Pak Angels Generative & Agentic AI Training - Cohort 11: participated in the mid-program hackathon and Final Hackathon 2; extended WatchMyWork with an Adaptive Recovery/Search Agent.
- Aspire Leaders Program - Alumni.

CERTIFICATIONS & ADDITIONAL TRAINING

AI / Automation: Build Intelligent Agents Using DeepSeek & n8n; ChatGPT + Zapier; Automating Sales Tasks with Zapier
Programming / Engineering: Python; Python Interaction with Operating System; MATLAB; Control Design; Business Analysis Fundamentals

LANGUAGES

English - Professional Proficiency | Urdu - Native | Saraiki - Native